

Manav Garg

Full Stack Software Development Engineer

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Professional Summary

Full-Stack Software Engineer specializing in **Machine Learning, Deep Learning,** and **Data Science**, shipping end-to-end from research to production. 2+ years across MNCs and startups, with a track record of **80%+ pipeline runtime cuts**, **70% hallucination reduction**, and **25% infra cost savings**. I deploy, monitor, optimize, and own the entire stack; building toward AI-native products at scale.

Professional Experience

Vault22

AI/ML Engineer, Contract
Dubai/South Africa, Remote

October 2025 – April 2026

- Cut infrastructure and ML pipeline costs by **22%** via **SageMaker** optimization and workload restructuring. Deployed self-hosted model serving on **AWS** with **Docker** and **Kubernetes**, raising inference reliability by **40%** and cutting compute overhead by **30%**.
- Rebuilt **ETL pipelines** and the data lake from scratch, lifting data integrity by **60%**. Shipped real-time **QuickSight** KPI dashboards for leadership visibility, improving report turnaround by **75%**; optimized **MongoDB** indexes, eliminating **95%** of query errors.

IHX Pvt. Ltd. (A Perfios Company)

Associate Data Scientist
Bangalore, Onsite

January 2025 – March 2026

- Engineered **end-to-end ML pipelines** for document digitization at **92% accuracy**, cutting processing time from over **30 hours** to under **5 hours** and improving data quality by **35%**.
- Built **LangChain** pipelines that reduced **LLM hallucination** by over **70%** with zero fine-tuning, and an **async inference system** serving over **10,000 daily requests** at under **200 ms** latency. Set shared tooling standards with internal **Python** libraries adopted across **3 teams**, cutting iteration time by **40%**.
- Researched multimodal **RAG** architectures with **YOLO**, **Donut**, and custom encoder-decoders for document understanding.

Intel Corporation

AI/ML Intern
Chennai, Onsite

May 2024 – August 2024

- Delivered an **end-to-end NLP pipeline** using **HuggingFace Transformers** and event-driven architecture to classify contract clauses, detect deviations, and flag fraud at **94% accuracy**.
- Applied semantically rich **embeddings** to extract fine-print signals for contract validation, delivering **89% precision** in anomaly detection for legal and compliance analysis.

Tisac Pvt. Ltd.

Full Stack & ML Intern
Mumbai, Remote

May 2024 – June 2024

- Deployed enterprise-grade **Jupyter Docker Lab** environments with **Kubernetes** orchestration and a custom **Docker API** service, cutting environment provisioning time by **90%**.
- Configured **TLS/SSL HTTPS** subdomains per container, reverse proxies, **Firestore Auth**, and **MongoDB** storage, delivering a fully secure platform from scratch.

Hogarth India (WPP)

Full Stack Developer Intern
Gurugram, Onsite

August 2023 – October 2023

- Delivered live features for the Coca-Cola and Thums Up ICC World Cup 2023 campaign, coordinating with brand stakeholders on vote logging, code redemption, and real-time polling in **C#**, **.NET**, and **React**, serving over **500,000 user interactions**.

Wipro

Android Developer & ML Intern
Chennai, Onsite

August 2023 – November 2023

- Built an **ML-powered Android** application for Wipro Infrastructure Engineering, using **machine vision** to detect and classify pathway obstructions for autonomous rover navigation at **97% accuracy**.

Education

Vellore Institute of Technology, Chennai

B.Tech, Computer Science Engineering | GPA: 8.51 out of 10

August 2021 – June 2025

Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, XAI, Linear Algebra, Multivariate Calculus, Operating Systems, Computer Networks, Probability & Statistics, OOP

Projects

1. Orqestra - Self-Hosted AI Orchestration Platform

Cloud-run multi-agent workloads are costly and opaque mid-run. Built a self-hosted platform that hosts LLMs locally, sandboxes agent code, and streams live activity; **Go**, **Python**, **Docker**, **Redis**, **WebSockets**, with **AgentHive** and **LiteLLM** over an OpenAI-compatible API.

2. LuminSkin - Skin Cancer Classification via Vision Transformers

Skin cancer screening is limited by specialist access. Trained and benchmarked **ViT** and **DeepViT** in **PyTorch** with transfer learning, reaching **95%+ accuracy** on lesion classification.

3. Satellite Image Analysis - Vision Transformers

Land-use labeling is hard to scale manually. Benchmarked **ViT** against **Compact Convolutional Transformer (CCT)** to automate it, hitting **96% accuracy** across land-use categories.

4. KeyKeg - Password Strength Analyzer and Manager

Rule-based password meters miss real weakness patterns. Built a full-stack manager that scores strength, stores credentials securely, and generates passwords; a **Random Forest** model reached **93% accuracy**.

5. NSCHR - Web HMIS Portal

Clinics lost time to paper-based records. Designed and shipped a full-stack Health Management Information System (HMIS) that cut retrieval time **threefold**; sold to clinics in production.

Technical Skills

Programming Languages

Python, Go, C/C++, JavaScript/TypeScript, Java, SQL, Bash

Machine Learning & AI

PyTorch, TensorFlow, Scikit-Learn, HuggingFace, Computer Vision, NLP, Vision Transformers (ViT), YOLO, OpenCV, Document AI (Donut)

LLM & Generative AI

LangChain, LangGraph, LangSmith, RAG, Generative AI, Prompt Engineering

MLOps & Cloud

CI/CD, Model Deployment, Containerization (Docker, Kubernetes), Cloud Computing (AWS: EC2, S3, SageMaker), ETL Pipelines, Jenkins

Backend & Web

FastAPI, Django, Node.js/Express, React/Next.js, REST APIs

Databases & Tools

MongoDB, Firebase, Git, Linux, VS Code

Professional Competencies

System Architecture, Technical Leadership, Product Ownership, Stakeholder Management, Cross-Functional Execution, Performance Optimization, Production Operations, Strategic Planning

Certifications

Imperial College of London

Mathematics for Machine Learning: Linear Algebra
Credential ID 6SXR88VFWQKG

May 2026